

Exact Gap Doubling for Harmonic Langevin Mori–Zwanzig Orthogonal Dynamics

Candidate solution packet for Zhu, arXiv:2109.01990

agent_lane_03 Model GPT5.6

26 August 2026

Status: candidate full result, likely valid. For a natural rank-one Mori projection in a genuinely degenerate harmonic Langevin system, the optimal positive orthogonal decay gap is exactly twice the original gap. The fluctuation force and memory kernel have an even faster exact exponential law.

1 Source question and convention repair

Zhu’s Section 5, open problem 1, asks whether the Mori–Zwanzig orthogonal dynamics can rigorously have a larger spectral gap and decay faster for a degenerate stochastic system [1]. The exact source passage is on PDF page 10. The source writes the desired inequality as $\lambda_Q < \lambda$, while its earlier estimates have the form $Ce^{-\lambda t}$. With positive decay exponents, faster decay means the reverse inequality. We therefore make the convention explicit. For a semigroup $U(t)$ on an invariant Hilbert space E , set

$$\text{gap}(U; E) := \sup\{\omega \geq 0 : \exists C < \infty \text{ with } \|U(t)|_E\| \leq Ce^{-\omega t} \ \forall t \geq 0\}.$$

Thus a larger positive gap means faster decay. In the signed spectral-edge convention, faster decay means that the rightmost nonzero spectral edge moves to the left. The theorem below proves both equivalent statements.

imaginary eigenvalues, i.e. $\sigma(\mathcal{Q}\mathcal{K}\mathcal{Q}) \cap i\mathbb{R} = \{0\}$. This is a simple task for operator $\mathcal{K}$ [14], while non-trivial for operator $\mathcal{Q}\mathcal{K}\mathcal{Q}$ essentially because the kernel of $\mathcal{Q}\mathcal{K}\mathcal{Q}$ is multi-dimensional and strongly depends on the form of projection operator $\mathcal{P}$. More discussions in this regard can be found in [48].

5. Conclusions and open problems

In this paper, we provide a simple but powerful hypocoercivity method to analyze the dynamical properties of the Mori-type, effective Mori-Zwanzig equation (EMZ) corresponding to degenerate stochastic systems. Our analysis shows that under the same conditions that lead to the geometric ergodicity of $e^{t\mathcal{K}}$, the orthogonal semigroup $e^{t\mathcal{Q}\mathcal{K}\mathcal{Q}}$ is also geometrically ergodic. As a result, the EMZ memory kernel and the fluctuation force decay to their equilibrium states exponentially fast, which confirm similar conclusions previously obtained by the hypoellipticity method [48] and pre-assumed in the reduced-order modeling [22, 24, 41]. The presented method is developed based on the elegant hypocoercivity framework introduced by Dolbeault, Mouhot and Schmeiser [4] and extended by Grothaus and Stilgenbauer [12, 13]. In particular, we note that the proof becomes direct once we identify a particular, invariant subspace $\mathcal{Q}H \cap \text{Ker}(\mathcal{K})^\perp$ and perform hypocoercivity analysis therein. The method is expected to apply to any degenerate stochastic systems where the DMS-GS hypocoercivity framework works [12, 11]. At last, we propose several interesting open problems on the analysis of the EMZ equation:

1. Is it possible to get $\lambda_{\mathcal{Q}} < \lambda$? Through the hypoellipticity and the hypocoercivity analysis, we can only get a convergence rate $\lambda_{\mathcal{Q}}$ for semigroup $e^{t\mathcal{Q}\mathcal{K}\mathcal{Q}}$ larger than λ (Compare (22) with (18)). However, numerical simulation results [44, 22, 18] for molecular systems often suggest the opposite, i.e. $\lambda_{\mathcal{Q}} < \lambda$, which means $\mathcal{Q}\mathcal{K}\mathcal{Q}$ has a larger spectrum gap and the memory kernel decays in a rate faster than the original dynamics. In fact, this is one of the main motivations to use the Mori-Zwanzig equation to do reduced-order modelling. Is it possible to prove $\lambda_{\mathcal{Q}} < \lambda$ rigorously for a certain degenerate stochastic system⁵?
2. What if $\mathcal{P}$ is an infinite-rank projection operator? Currently, we only developed the hypoellipticity and hypocoercivity methods that work for finite-rank projection operators. Another frequently used projection operator, Zwanzig-type operator [50, 3, 2, 16], is an infinite-rank projection operator [43, 2]. It is commonly assumed in the coarse-grained modeling that $e^{t\mathcal{Q}\mathcal{K}\mathcal{Q}}$ is also geometrically ergodic under the Zwanzig-type projection, but how to prove it rigorously?
3. What if $V(q)$ is singular? Can we get similar results for singular potential energy $V(q)$ such as the frequently used Lennard-Jones potential?
4. Can we use the stochastic analysis framework to prove the geometric ergodicity of $e^{t\mathcal{Q}\mathcal{K}\mathcal{Q}}$, such as the ones used in [27, 28, 34]?

⁵For non-degenerate stochastic systems, this might be a simpler task. Consider the d -dimensional quantum harmonic oscillator [49] with λ -coercive generator $\mathcal{D} = \mathcal{X}^* \mathcal{X}$, we have $\langle \mathcal{Q}\mathcal{D}\mathcal{Q}f, f \rangle = \langle \mathcal{D}\mathcal{Q}f, \mathcal{Q}f \rangle \geq \lambda_{\mathcal{Q}} \|\mathcal{Q}f\|^2 = \lambda_{\mathcal{Q}} \|f\|^2$ for $f \in \mathcal{Q}H \cap \text{Ker}(\mathcal{X})^\perp$. Since $\mathcal{Q}H \cap \text{Ker}(\mathcal{X})^\perp \subset \text{Ker}(\mathcal{X})^\perp \subset H$, we have $\lambda_{\mathcal{Q}} \leq \lambda$.

Source evidence: Section 5 and all of open problem 1, including its convention-revealing footnote, on the original PDF page 10.

2 Model and full result

Fix $\gamma > 2$ and consider the one-dimensional harmonic Langevin equation

$$dq_t = p_t dt, \quad dp_t = -q_t dt - \gamma p_t dt + \sqrt{2\gamma} dW_t. \quad (1)$$

Its backward generator is

$$\mathcal{K} = p\partial_q - q\partial_p - \gamma p\partial_p + \gamma\partial_p^2 \quad (2)$$

on $H = L^2(\mu)$, where $d\mu = (2\pi)^{-1} e^{-(q^2+p^2)/2} dq dp$. The diffusion in (1) has rank one in a two-dimensional phase space, so the system is genuinely degenerate. Its drift and noise nevertheless satisfy the Kalman condition.

Let $u(q, p) = q$. Since $\|q\|_H = 1$, the natural scalar Mori projection and its complement are

$$\mathcal{P}f = \langle f, q \rangle_H q, \quad \mathcal{Q} = I - \mathcal{P}.$$

Write $T(t) = e^{t\mathcal{K}}$ and let $T_{\mathcal{Q}}(t)$ be the orthogonal semigroup generated by the closure of $\mathcal{Q}\mathcal{K}\mathcal{Q}$ on $E = \mathcal{Q}H \cap \{1\}^\perp$. Finally put

$$a = \frac{\gamma - \sqrt{\gamma^2 - 4}}{2}, \quad b = \frac{\gamma + \sqrt{\gamma^2 - 4}}{2}. \quad (3)$$

Then $0 < a < b$, $a + b = \gamma$, and $ab = 1$.

Theorem 1 (Exact acceleration). *For the degenerate stochastic system (1),*

$$\begin{aligned} \text{gap}(T; \{1\}^\perp) &= a, \\ \text{gap}(T_{\mathcal{Q}}; E) &= 2a. \end{aligned}$$

Equivalently, the rightmost nonzero spectral edge moves from $-a$ for $\mathcal{K}$ to $-2a$ for $\mathcal{Q}\mathcal{K}\mathcal{Q}$. In particular, the positive orthogonal gap is strictly larger and in fact exactly doubles.

Moreover, the Mori fluctuation force $F(t)$ and scalar memory kernel $M(t)$ for $u = q$ are

$$F(t) = T_{\mathcal{Q}}(t)\mathcal{Q}\mathcal{K}q = e^{-\gamma t}p, \quad M(t) = \langle q, \mathcal{K}F(t) \rangle_H = -e^{-\gamma t}. \quad (4)$$

Thus these two quantities decay at the still faster rate $\gamma > 2a$.

3 Proof intuition

The Gaussian L^2 space splits orthogonally into Wiener chaoses. Harmonic Langevin evolution preserves every chaos, and the n th-chaos semigroup is the n th symmetric tensor power of the two-dimensional first-chaos semigroup. Consequently the slowest centered mode occurs in the first chaos.

The Mori projection onto q deletes one direction from that first chaos. The remaining first-chaos direction p is driven by the compressed scalar generator $-\gamma$. Crucially, the projection is zero on every higher chaos, so the slowest untouched mode is the tensor square of the original slow mode. Its exponent is exactly $2a$. This explains both the doubling and its sharpness; no hypocoercive estimate is being compared to another nonsharp estimate.

4 Proof

Let

$$B = \begin{pmatrix} 0 & 1 \\ -1 & -\gamma \end{pmatrix}, \quad e_p = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Then (1) is $dX_t = BX_t dt + \sqrt{2\gamma} e_p dW_t$. The identity

$$B + B^\top + 2\gamma e_p e_p^\top = 0$$

proves invariance of the standard Gaussian. Also e_p and $Be_p = (1, -\gamma)^\top$ span $\mathbb{R}^2$, proving the Kalman condition.

Let H_n be the n th Gaussian Wiener chaos, so $H = \bigoplus_{n \geq 0} H_n$, $H_0 = \text{span}\{1\}$, and $H_1 = \text{span}\{q, p\}$. We next derive, rather than assume, the semigroup action on these spaces.

Lemma 1 (Exact chaos action). *Set $S(t) = e^{tB^\top}$ on $\mathbb{R}^2 \cong H_1$. Then*

$$T(t)|_{H_n} = S(t)^{\odot n}, \quad \|T(t)|_{H_n}\| = \|S(t)\|^n \quad (n \geq 1),$$

where $\odot n$ denotes the n th symmetric tensor power.

Proof. For $z \in \mathbb{R}^2$, introduce the Wick exponential

$$\mathcal{E}_z(x) = \exp\left(z \cdot x - \frac{|z|^2}{2}\right).$$

Conditionally on $X_0 = x$, $X_t = e^{tB}x + Y_t$, where Y_t is centered Gaussian with covariance $I - e^{tB}e^{tB^\top}$. Gaussian integration gives

$$T(t)\mathcal{E}_z(x) = \exp\left((S(t)z) \cdot x - \frac{|S(t)z|^2}{2}\right) = \mathcal{E}_{S(t)z}(x).$$

Comparing homogeneous terms in z proves the symmetric-tensor identity. The upper norm bound is the ordinary tensor-power bound. Equality follows by choosing a unit vector at which the two-dimensional operator $S(t)$ attains its norm and taking its symmetric pure tensor. $\square$

Because $T(t)$ is a Markov contraction, $\|S(t)\| \leq 1$. Lemma 1 therefore yields the exact centered norm identity

$$\|T(t)|_{\{1\}^\perp}\| = \sup_{n \geq 1} \|S(t)\|^n = \|S(t)\|. \quad (5)$$

On the ordered basis (q, p) , the first-chaos generator is

$$\mathcal{K}|_{\mathbf{H}_1} = B^\top = \begin{pmatrix} 0 & -1 \\ 1 & -\gamma \end{pmatrix}, \quad \mathcal{K}q = p, \quad \mathcal{K}p = -q - \gamma p. \quad (6)$$

Its two distinct eigenvalues are $-a$ and $-b$. Finite-dimensional diagonalization gives constants $0 < c \leq C < \infty$ with

$$ce^{-at} \leq \|S(t)\| \leq Ce^{-at} \quad (t \geq 0).$$

Together with (5), this proves that the original sharp gap is a .

Now use that $\mathcal{P}$ has range in $\mathbf{H}_1$. Since $\mathcal{K}$ preserves every chaos, $\mathcal{Q}$ is the identity and $\mathcal{Q}\mathcal{K}\mathcal{Q} = \mathcal{K}$ on $\mathbf{H}_n$ for every $n \geq 2$. Within the first chaos, $\mathcal{Q}\mathbf{H}_1 = \text{span}\{p\}$ and (6) gives

$$\mathcal{Q}\mathcal{K}\mathcal{Q}p = -\gamma p.$$

Consequently

$$E = \text{span}\{p\} \oplus \bigoplus_{n \geq 2} \mathbf{H}_n, \quad T_{\mathcal{Q}}(t) = e^{-\gamma t} I_{\text{span}\{p\}} \oplus \bigoplus_{n \geq 2} T(t)|_{\mathbf{H}_n}. \quad (7)$$

Polynomial chaoses form a core for the Gaussian Ornstein–Uhlenbeck generator. Moreover $\langle \mathcal{K}f, q \rangle = \langle f, \mathcal{K}^*q \rangle$ with $\mathcal{K}^*q = -p$, so the rank-one compression term extends boundedly to H . Thus the closure of the displayed block operator is precisely the orthogonal generator, justifying (7) at the semigroup level.

Applying Lemma 1 once more gives the exact norm formula

$$\|T_{\mathcal{Q}}(t)|_E\| = \max\{e^{-\gamma t}, \sup_{n \geq 2} \|S(t)\|^n\} = \max\{e^{-\gamma t}, \|S(t)\|^2\}. \quad (8)$$

Since $\gamma = a + b > 2a$, equations (8) and (5) imply upper and lower bounds of order e^{-2at} . Hence the orthogonal sharp gap is $2a$.

For completeness, the same calculation identifies the point spectral sets: the nonconstant eigenvalues of $\mathcal{K}$ are

$$\{-(ma + nb) : m, n \in \mathbb{N}_0, m + n \geq 1\},$$

whereas after compression the first-chaos slow mode is gone and the retained nonzero values are

$$\{-\gamma\} \cup \{-(ma + nb) : m, n \in \mathbb{N}_0, m + n \geq 2\}.$$

The rightmost elements are respectively $-a$ and $-2a$.

Finally, (6) gives $\mathcal{Q}\mathcal{K}q = p$, and (7) gives $T_{\mathcal{Q}}(t)p = e^{-\gamma t}p$. Since $\langle q, p \rangle = 0$ and $\|q\| = 1$,

$$\langle q, \mathcal{K}T_{\mathcal{Q}}(t)p \rangle = e^{-\gamma t} \langle q, -q - \gamma p \rangle = -e^{-\gamma t}.$$

This proves (4) and completes the theorem. $\square$

5 Verification, novelty, and limitations

The included checker constructs $\mathcal{K}$ on the normalized product-Hermite basis through total degree ten. For $\gamma \in \{2.1, 2.5, 3, 5, 10\}$ it verifies all eigenvalues $-(ma + nb)$, the exact symmetric-tensor semigroup norms at three times, the compressed first-chaos block, and (4). All checks pass. These computations are regression tests; Lemma 1 and the direct-sum argument are the proof.

A bounded novelty audit searched the run’s lightweight indexes and the web by the exact paper id/title and close combinations of “orthogonal dynamics”, “harmonic/underdamped Langevin”, “spectral gap”, and $\lambda_{\mathcal{Q}}$. It also inspected the relevant parts of arXiv:2503.20457, 2604.20453, and 2607.25855. Those papers address semigroup well-posedness or infinite-rank Zwanzig projections; their harmonic examples do not state this stochastic rank-one gap-doubling theorem. No exact duplicate was found as of 26 August 2026. This is bounded evidence, not an exhaustive literature claim.

The theorem answers only the first open problem on source page 10, after repairing its sign convention. It does not treat the infinite-rank Zwanzig projection, singular potentials, or the requested stochastic-analysis proof. The restriction $\gamma > 2$ avoids the critically damped Jordan case and keeps the exact-rate statement free of polynomial prefactors.

Human review focus. Verify the Wick-exponential passage to symmetric tensor powers, the core/closure identification behind (7), and the intended interpretation of the source’s reversed inequality. Subject to that convention repair, the answer is complete.

References

- [1] Y. Zhu, *A simple hypocoercivity analysis for the effective Mori–Zwanzig equation*, arXiv:2109.01990, 2021.